

MEHER SONI

AI Researcher : Speech AI · Indic NLP · Knowledge Graphs · Trustworthy AI

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PROFILE

Final-year B.Tech AI researcher working on low-resource and multilingual Indic language technologies, including phoneme-level pronunciation modeling for Urdu, Punjabi and Indian English (IIT Ropar), noise-robust speech recognition (DRDO) and trustworthy AI systems that ground LLM responses in verifiable evidence. First author on a Springer-published paper that reduced hallucinated legal citations by 82% in an Indian legal AI system, with additional publications in knowledge graphs and recommender systems. Experienced across the full AI pipeline: from phonetics and label-set design to speech model training and grounded LLM systems.

RESEARCH & INDUSTRY EXPERIENCE

Research Intern - Multilingual Speech & Language Technology · IIT Ropar

May 2026 – Jul 2026

- Developed the **Urdu front-end** for a **multilingual Indic TTS framework** by designing phoneme inventories, Common Label Set (CLS) mappings and pronunciation rules from scratch, alongside parallel development for Punjabi and Indian English.
- Designed context-aware grapheme-to-phoneme rules for certain ambiguous Urdu characters, improving pronunciation accuracy in multilingual text-to-speech systems.

R&D Intern - Speech Recognition · DRDO - Defence Electronics Applications Lab (DEAL), Dehradun

Jun 2025 – Jul 2025

- Designed and benchmarked hybrid CNN-LSTM architectures for **noise-robust automatic speech recognition (ASR)** on a defence-oriented speech-to-text project focused on acoustic recognition in adverse conditions.

R&D Intern · IIIT Vadodara

Mar 2025 – Jul 2025

- Redesigned the conversational recommendation pipeline using GraphSAGE, improving **Recall@100 by 155% (0.0680 → 0.1734)**.
- Improved generation diversity (Distinct-4: 0.0764), measurably reducing repetitive model outputs in generated recommendations.

R&D Intern · Svaarogyam Medical Devices Pvt. Ltd.

Jul 2024 – Jan 2025

- Led **non-invasive diabetes-detection** research across ~500 patient samples using wearable optical/infrared sensors, from data collection design through model validation.
- Delivered an SVM model (83% accuracy) and a functional Arduino prototype for real-time edge inference.

ML Intern · MNIT Jaipur & Svaarogyam Medical Devices Pvt. Ltd.

May 2024 – Jul 2024

- Built, evaluated multi-disease prediction pipeline (Alzheimer's 91.4%, Diabetes 86.1%, Heart Disease 88.7%) using RF, SVM, XGBoost.
- Owned clinical data preprocessing and supported hardware prototype development for healthcare deployment.

PUBLICATIONS & CONFERENCES

Reducing Hallucinated Legal Citations in LLM Responses for Indian Law - First Author, IDBA 2026, Springer (6th Intl. Conf. on Data Science & Big Data Analytics, ACM Indore Chapter.)

- Built a system that checks an AI model's legal citations before they reach the user using citation-aware knowledge graph verification.
- Cut false citations by **82% (CHR: 0.343 → 0.063)** and reached a Retrieval Reliability Score of 0.925.

Knowledge Graph Construction from Review-Based Conversational Data - First Author, HACER 2025, IIIT Vadodara

- Engineered a Neo4j-based KG pipeline with entity-relation extraction and metadata enrichment
- Concluded inverse relations improve multi-hop retrieval efficiency.

Explainable Drug Adoption Forecasting using Knowledge Graphs - Co-Author, IDBA 2026, Springer (Learning and Analytics in Intelligent Systems series)

- Developed a knowledge-graph-driven forecasting model combining explainability with predictive power for drug adoption.

KEY PROJECTS

Speech-to-Text via Deep Learning

[GitHub](#)

- Trained CNN-BiLSTM and Wav2Vec 2.0 ASR models using FFT/Mel-spectrogram features, CTC loss and Beam Search decoding with an N-gram language model.
- Improved transcription accuracy using **Beam Search decoding** with an **N-gram language model**, achieving **6.43% WER** on the **LibriSpeech** dataset.

Nyaya.AI - Graph-Constrained Legal Reasoning

[Live Demo](#) | [GitHub](#)

- Developed a **GraphRAG-based legal AI** integrating Knowledge Graphs and LLMs for grounded, citation-backed legal question answering.
- **Manually curated a legal dataset** from India Code statutes and structured statutory provisions for Knowledge Graph construction, retrieval and grounded legal reasoning.

Safety Smart Wristband - SUDARSHANA

[GitHub](#)

- Developed a **Random Forest-based panic detection model** using multimodal physiological signals (heart rate, GSR, temperature and accelerometer data), achieving **99.7% F1-score** for distress classification.
- Designed an **AI-powered IoT wearable** integrating Arduino, GPS, GSM, heart rate and GSR sensors for autonomous SOS triggering and real-time emergency response.

TECHNICAL SKILLS

- **Speech & Multilingual NLP:** Pronunciation-rule design, phoneme inventories (CLS/ILSL12), text-to-speech and speech-recognition pipelines, wav2vec2.0, multi-script text processing (Urdu, Punjabi, Hindi, Indian English)
- **AI Systems & Retrieval:** Retrieval-augmented generation, grounding AI answers in real sources, prompt engineering
- **Knowledge Graphs:** Neo4j, Cypher, Py2Neo, GraphSAGE, entity-relation extraction, multi-hop graph traversal
- **ML / DL:** PyTorch-based CNN-LSTM, XGBoost, SVM, Random Forest, feature engineering
- **Languages & Data:** Python (NumPy, Pandas, Scikit-learn), SQL, C (Embedded)
- **Tools:** Git, Arduino, ESP-LilyGo (IoT/Embedded), Streamlit

EDUCATION

Birla Institute of Technology, Mesra, Jaipur - B.Tech, Artificial Intelligence · SGPA 9.62/10 2023–2027
Coursework: Machine Learning, Deep Learning, Feature Engineering, Probability & Statistics, DBMS, Image Processing, Information Retrieval

Indian Institute of Technology, Madras - B.S., Data Science · CGPA 7.5/10 2024–2028
Coursework: ML Techniques, Business Data Management, ML Foundations, Python, Statistics, Tools for Data Science

ACHIEVEMENTS & DISTINCTIONS

Certifications:

GATE DA 2026 Qualified • Amazon ML Summer School 2026 • Prompt Engineering (Vanderbilt) • AI/ML for Geodata Analysis (ISRO–IIRS)
• Joy of Computing (NPTEL–IIT Madras) • Big Data, AI & Ethics (UC Davis) • Google UX Design

Achievements:

Renowned Shooter- 68th National Shooting Championship (10m Air Pistol) • Certified Skier (IISM Gulmarg)